Double Homodyne Detection of Squeezed Vacuum and Coherent Light for Quantum Interferometry

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Abstract

Quantum interferometry techniques are a prominent emerging quantum technology with uses which range from bioimaging to the detection of gravitational waves. By interfering quantum states of light to probe some parameter, encoded into its phase, one can increase the fundamental precision achievable by a measurement. These techniques often rely on exotic quantum states and sophisticated measurements performed by expensive technologies still in their infancy, such as photon number-resolving detectors. This thesis presents a phase estimation scheme which uses coherent and squeezed light mixed on a beamsplitter to encode phase information, and subsequently measures the phase using two homodyne detectors. We demonstrate that this scheme saturates the quantum Cramér-Rao bound, which achieves Heisenberg scaling. We further show that even in the presence of substantial photon loss, the proposed detection scheme still beats the standard quantum limit and requires no adjustments depending on the true value of the phase to be measured. Compared many alternatives, the proposed scheme is relatively practical to realize, as its two essential components, squeezed vacuum state preparation and homodyne detection, are already widespread and well-established techniques in quantum optics.

I. INTRODUCTION

Phase estimation in interferometry, that is, the determination of a phase difference ϕ between interfering modes of light, is a problem of scientific interest in various fields of physics. In standard interferometry, using classical states of light, the variance in the phase measurement scales inversely with the average photon number $\Delta\phi^2 \propto 1/\bar{n}$. This sets a benchmark, referred to as the standard quantum limit (SQL), to surpass using quantum states of light. In certain applications high intensity (high $\bar{n}$) light cannot be used, such as in gravitational wave detection [1–3], where intense light can introduce vibrations to a sensitive setup, or in bioimaging [4, 5], where biological matter might be damaged. These applications require high precision despite being constrained in the average photon number. To achieve this, it has been shown that certain quantum states can be used to achieve better scaling between the precision and the photon number [6]. Many of these techniques, however, require highly sophisticated state preparation and state measurement setups. A recent work from McIntyre and Coish [7] has demonstrated that using entangled coherent states (ECS) measured with combined homodyne detection gives the optimal uncertainty scaling achievable as a result of the uncertainty principle – $\Delta\phi^2 \propto 1/\bar{n}^2$ (Heisenberg scaling) – making it applicable in limited-intensity regimes. Homodyne detection is a highly accessible measurement, not

requiring the sophisticated equipment needed for other alternative measurements, such as number state resolving detectors. Like most states for quantum interferometry, the ECS is very difficult to realize experimentally. For low photon number, however, the ECS can be approximated by the so-called Caves state [8], first studied for interferometry in a seminal work by Dr. Carleton M. Caves [9], achieved by mixing a coherent state with a squeezed vacuum state on a beam splitter. This project reproduces the calculation in [7] with the Caves state to determine its utility for quantum interferometry.

II. THEORETICAL BACKGROUND

A. Quantum Metrology

1. Probability in Quantum Mechanics

The feature of quantum mechanics which most distinguishes it from a classical theory is its fundamentally probabilistic, or non-deterministic, nature: Quantum mechanics can, at best, predict the probabilities associated with different measurement outcomes. This indeterminism is not the result of our ignorance about a physical system, nor a consequence of an unideal measurement, but seems to be a fundamental feature of reality. Unfortunately, this places fundamental limits on the precision of our measurements. However, by understanding the nature of this uncertainty and carefully designing our measurement scheme, we can improve upon standard metrology techniques and achieve new heights of precision.

In the phase estimation problem, in which some fixed phase ϕ is to be measured by probing with photon number measurements, the precision of the measurement is fundamentally limited by so-called Heisenberg limit: $\Delta\phi \propto 1/N^2$ [10]. This arises from the generalized Heisenberg uncertainty principle, as the phase and particle number, being conjugate variables, are mutually incompatible observables. Achieving the Heisenberg limit is therefore the gold standard for phase estimation.

2. Fisher Information and the Cramér-Rao Bound

The classical Fisher information, often simply called Fisher information, is a measure of the usefulness of a measurement outcome in determining the value of some parameter. Given a probability density function (PDF) $p(x|\phi)$, which gives the probability of the measurement M having the outcome x which depends on the parameter ϕ, the classical Fisher information

is given by the integral:

$$\begin{aligned} FI(\phi\,;M) &= \int_{\mathcal{S}} dx\; p(x|\phi)\left(\frac{\partial}{\partial\phi}\log(p(x|\phi))\right)^2, \\ &= \int_{\mathcal{S}} dx\; p(x|\phi)^{-1}\left(\frac{\partial}{\partial\phi}p(x|\phi)\right)^2, \end{aligned} \tag{1}$$

where $\mathcal{S}$ is the set of possible outcomes x. The Fisher information quantifies the sensitivity of $p(x|\phi)$ to the value of ϕ. In this sense, for M to be a strong estimator of ϕ, the associated PDF $p(x|\phi)$ ought to be highly sensitive to ϕ, and thus ought to have a high Fisher information. This fact is made explicit by the so-called Cramér-Rao bound, which states that the statistical variance in estimating ϕ via the measurement M, is at minimum the reciprocal of the classical Fisher information [11]:

$$\Delta\phi^2 \geq \frac{1}{FI(\phi\,;M)}. \tag{2}$$

This fundamental limit on the precision has important consequences for quantum information theory, since measurements of quantum states are, by their very nature, probabilistic. A common task in quantum information theory is to find measurements which maximize the classical information. Though finding the corresponding measurement is non-trivial, the value of the maximum classical Fisher information can be calculated without knowledge of any measurement which achieves it. This maximum is known as the quantum Fisher information. Given an encoding unitary transformation defined by a generator A, on a pure state $|\psi\rangle$:

$$U_\phi = \exp(-iA\phi), \tag{3}$$

$$|\psi\rangle \to |\psi_\phi\rangle = U_\phi|\psi\rangle, \tag{4}$$

the quantum Fisher information is four times the variance of A with respect to $|\psi\rangle$ [12]:

$$\begin{aligned} FI_Q(\phi\,;A) &= 4\,\mathrm{Var}(A) \\ &= 4\,(\langle\psi|A^2|\psi\rangle - \langle\psi|A|\psi\rangle^2). \end{aligned} \tag{5}$$

In contrast to the classical Fisher information, the quantum Fisher information is independent of any measurement made, but depends only on the parameter ϕ and its relation to the quantum state $|\psi_\phi\rangle$ which carries information about it. In fact, the quantum Fisher information is the supremum of the classical Fisher information for any possible measurement M which estimates ϕ [13]:

$$FI_Q(\phi\,;A) = \sup_M FI(\phi\,;M). \tag{6}$$

B. Quantum Optics

1. *Electromagnetic Field Quantization*

To discuss enhanced interferometry using quantum states, one of course needs a suitable quantum theory of light. We turn to Walls and Milburn's [14] discussion of the quantization of the electromagnetic (EM) field. We consider the Coulomb gauge, imposing that the vector potential be solenoidal: $\nabla \cdot \mathbf{A} = 0$. This gives $\mathbf{B} = \nabla \times \mathbf{A}$ and $\mathbf{E} = -\frac{\partial \mathbf{A}}{\partial t}$. The Ampere-Maxwell law (in the absence of any charges or currents) becomes:

$$\nabla \times \mathbf{B} = \frac{1}{c^2}\frac{\partial \mathbf{E}}{\partial t} \tag{7}$$

$$\nabla \times (\nabla \times \mathbf{A}) = -\frac{1}{c^2}\frac{\partial^2 \mathbf{A}}{\partial t^2} \tag{8}$$

$$\nabla^2 \mathbf{A} = \frac{1}{c^2}\frac{\partial^2 \mathbf{A}}{\partial t^2} \tag{9}$$

where in the last step we have used the vector identity $\nabla \times (\nabla \times \mathbf{V}) = \nabla(\nabla \cdot \mathbf{V}) - \nabla^2 \mathbf{V}$. Thus $\mathbf{A}$ is a classical wave. A general solution is written as:

$$\mathbf{A}(\mathbf{r}, t) = \sum_k \frac{\hbar}{2\omega_k \epsilon_0} \left[c_k \mathbf{u}_k(\mathbf{r}) \mathrm{e}^{-i\omega_k t} + c_k^* \mathbf{u}_k^*(\mathbf{r}) \mathrm{e}^{i\omega_k t} \right] \tag{10}$$

where $\mathbf{u}_k(\mathbf{r})$ are orthogonal mode vectors, labeled by k, satisfying the Poisson equation: $\left(\nabla^2 + \frac{\omega_k^2}{c^2}\right) \mathbf{u}_k(\mathbf{r}) = 0$ (dependent on boundary conditions), and c_k are dimensionless complex-valued coefficients. The quantization of the field is performed by by swapping these coefficients for operators $\hat{a}_k, \hat{a}_k^\dagger$ satisfying the bosonic commutation relations: $[\hat{a}_k, \hat{a}_{k'}^\dagger] = \delta_{k,k'}$ and $[\hat{a}_k, \hat{a}_{k'}] = [\hat{a}_k^\dagger, \hat{a}_{k'}^\dagger] = 0$. We now have the vector potential field operator:

$$\hat{\mathbf{A}}(\mathbf{r}, t) = \sum_k \frac{\hbar}{2\omega_k \epsilon_0} \left[\hat{a}_k \mathbf{u}_k(\mathbf{r}) \mathrm{e}^{-i\omega_k t} + \hat{a}_k^\dagger \mathbf{u}_k^*(\mathbf{r}) \mathrm{e}^{i\omega_k t} \right] \tag{11}$$

which can be used to find the electric and magnetic field operators via $\hat{\mathbf{B}} = \nabla \times \hat{\mathbf{A}}$ and $\hat{\mathbf{E}} = -\frac{\partial \hat{\mathbf{A}}}{\partial t}$. Finally, we consider the Hamiltonian of the EM field, which can be shown to simplify as follows:

$$\hat{H} = \int dV \left(\frac{\hat{\mathbf{B}}^2}{2\mu_0} + \frac{\epsilon_0 \hat{\mathbf{E}}^2}{2} \right) \tag{12}$$

$$= \sum_k \hbar\omega_k \left(\hat{a}_k^\dagger \hat{a}_k + \frac{1}{2} \right). \tag{13}$$

The above equations reveal that the EM field can be treated as a quantum harmonic oscillator with canonically-conjugate variables $\hat{\mathbf{B}}$ and $\hat{\mathbf{E}}$. From Eq. (13), we can see that energy

eigenstates of the field are simultaneously eigenstates of the number operator in each mode $\hat{n}_k = \hat{a}_k^\dagger \hat{a}_k$:

$$\hat{n}_k |n_k\rangle = n_k |n_k\rangle \tag{14}$$

The excitations of the EM field are called photons, thus energy eigenstates of the field are states with defined number of photons.

2. Phase-Space Representations

In Sec. II B 1 we found that the EM field is described by a quantum harmonic oscillator with canonically-conjugate field operators $\hat{\mathbf{B}}$ and $\hat{\mathbf{E}}$. Like the canonically conjugate variables position x and momentum p, the field operators are analogous to phase variables and are subject to the Heisenburg uncertainty principle and are thus incompatible observables. This point is crucial to our understanding of the phase estimation problem. However, we need to treat a more general case of conjugate operators. We do this by defining the generalized quadrature operators $\hat{x}_\theta$ and $\hat{p}_\theta$:

$$\hat{x}_\theta = \frac{1}{2}(\mathrm{e}^{i\theta}\hat{a}^\dagger + \mathrm{e}^{-i\theta}\hat{a}) \tag{15}$$

$$\hat{p}_\theta = \frac{i}{2}(\mathrm{e}^{i\theta}\hat{a}^\dagger - \mathrm{e}^{-i\theta}\hat{a}). \tag{16}$$

By inverting this relationship, $\hat{x}_\theta$ and $\hat{p}_\theta$ can be understood as the real and imaginary components of the annihilation operators, rotated by some angle θ: $\hat{a} = \mathrm{e}^{i\theta}(\hat{x}_\theta + i\hat{p}_\theta)$. Their commutation relation reveals that they are canonically conjugate, leading to an associated Heisenberg uncertainty relation:

$$[\hat{x}_\theta, \hat{p}_\theta] = \frac{i}{2}, \tag{17}$$

$$\Delta\hat{x}_\theta \Delta\hat{p}_\theta \geq \frac{1}{4}. \tag{18}$$

So-called gaussian states will saturate this bound and are often of interest in quantum interferometry. The gaussian states include coherent and squeezed states, which we study for use in phase estimation in the following sections. Rewritten in terms of the quadrature operators, Eq. (13) becomes:

$$\hat{H} = \sum_k \hbar\omega_k \left(\hat{x}_{\theta,k}^2 + \hat{p}_{\theta,k}^2\right) \tag{19}$$

Any state of the EM field, described by its density matrix $\hat{\rho}$, can be represented in terms of the generalized quadrature operators via the Wigner function, defined as [14]:

$$W(x_\theta, p_\theta) = \frac{1}{\hbar\pi}\int_{\mathbb{R}} dy\, \mathrm{e}^{2iyp_\theta}\langle x_\theta + y|\hat{\rho}|x_\theta - y\rangle. \tag{20}$$

where $|x_\theta\rangle$ is an eigenstate of the $\hat{x}_\theta$ operator. The Wigner function is a quasi-probability distribution, and it is useful for visualizing the state $\hat{\rho}$ in phase space. The probability distribution for measuring the quadrature x_θ is given integrating over the conjugate variable:

$$P(x_\theta) = \int_{\mathbb{R}} dp_\theta \, W(x_\theta, p_\theta) \tag{21}$$

3. Coherent Light

One type of state which saturates the Heisenburg uncertainty lower bound given by Eq. (18) is known as a coherent state. They are, to a good approximation, the state which is produced by a laser [15]. They are defined as the eigenstates of the annihilation operator, labeled by the complex-valued eigenvalue α:

$$\hat{a}|\alpha\rangle = \alpha|\alpha\rangle \tag{22}$$

Knowing how the annihilation operator acts on the number states, one can easily find the number state basis representation of the coherent states to be:

$$|\alpha\rangle = \mathrm{e}^{-|\alpha|/2} \sum_n \frac{\alpha^n}{\sqrt{n!}} |n\rangle \tag{23}$$

which one will recognize as a Poissonian distribution. Since they are not eigenstates of the Hamiltonian, the coherent states evolve over time:

$$\begin{aligned} |\alpha(t)\rangle &= \mathrm{e}^{i\hat{H}t/\hbar}|\alpha\rangle \\ &= \mathrm{e}^{-|\alpha|/2} \sum_n \frac{\alpha^n}{\sqrt{n!}} \mathrm{e}^{i\omega n t}|n\rangle \\ &= \mathrm{e}^{-|\alpha|/2} \sum_n \frac{(\alpha \mathrm{e}^{i\omega t})^n}{\sqrt{n!}}|n\rangle \\ &= |\alpha \mathrm{e}^{i\omega t}\rangle \end{aligned} \tag{24}$$

where we have omitted the constant energy term in the Hamiltonian. We can see that coherent states evolve in time via a rotation in phase space: This property makes the coherent state useful in interferometry as it acquires a well-defined phase in time. To visualize the coherent state, we can use the Wigner transform from Eq. (20) [14]:

$$\begin{aligned} W_\alpha(x_\theta, p_\theta) &= \frac{1}{\pi} \int_{\mathbb{R}} dy \, \mathrm{e}^{2iyp_\theta} \langle x_\theta + y \,|\, \alpha\rangle\langle\alpha \,|\, x_\theta - y\rangle, \\ &= \frac{2}{\pi} \exp\left(-2(x_\theta - \mathrm{Re}[\alpha])^2 - 2(p_\theta - \mathrm{Im}[\alpha])^2\right). \end{aligned} \tag{25}$$

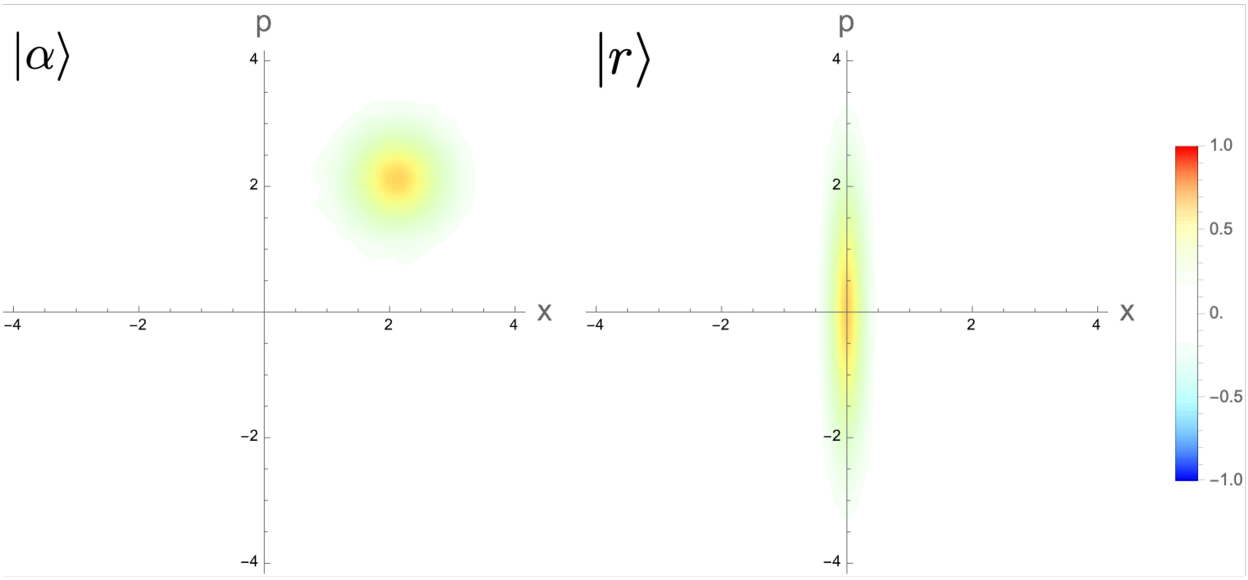


FIG. 1: Wigner distributions for (left) a coherent state with amplitude $\alpha = 3e^{i\pi/4}$ and (right) a squeezed vacuum state with $r = 1$.

Plotted in Fig. 1, the coherent state is represented as a 2D gaussian distribution in phase space with standard deviation $\sigma = \frac{1}{2}$ in both quadratures, saturating the Heisenberg uncertainty relation Eq. (18).

4. Squeezed Light

Similar to the coherent state is the squeezed state. The squeezed state is also represented by a minimally uncertain 2D gaussian in phase space. However, whereas the coherent state had equal variance in each of its quadratures $\sigma_x = \sigma_p = \frac{1}{2}$, the squeezed state has one "squeezed" quadrature with $\sigma_{x_\theta} = \frac{1}{2}e^{-r}$ and one "anti-squeezed" quadrature with $\sigma_{p_\theta} = \frac{1}{2}e^{r}$ for some real number r and complex phase θ. The complex number $\xi = re^{i\theta}$ parametrizes the squeezed states and is called the squeezing parameter. A squeezed state can also be displaced (like a coherent state), however the squeezing and displacement operations do not commute [14]. In this project, we will only consider the squeezed vacuum (non-displaced) state. The squeezed vacuum is visualized in Fig. 1, which the gaussian Wigner function [14] for the squeezed state:

$$W_r(x_\theta, p_\theta) = \frac{2}{\pi} \exp\left(-2e^{2r}x_\theta^2 - 2e^{-2r}p_\theta^2\right) \tag{26}$$

C. Quantum Interferometry

1. *The Mach-Zehnder Interferometer*

A common setup for use in interferometry is known as the Mach-Zehnder interferometer, depicted in Fig.2. It consists of two modes of light mixed on a beamsplitter, sent through two 'arms', each imparting a different phase ϕ_1 and ϕ_2 to each mode, before being mixing the modes on a second beamsplitter to be measured at the two output ports. Generally, the phase imparted to the light is used a proxy for the path length of each arm. This has a variety of applications, including gravitational wave detection [1, 2], and sensing/bioimaging [5].

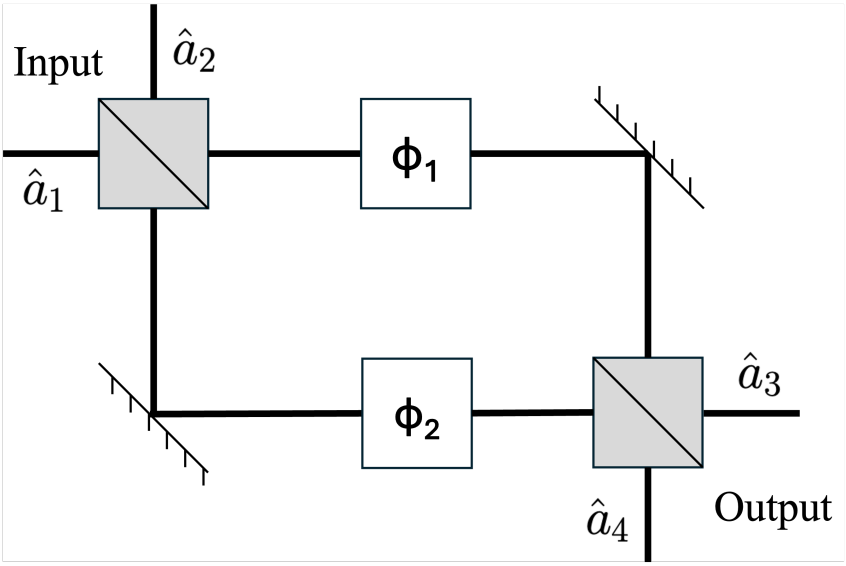


FIG. 2: A generic Mach-Zehnder interferometer, imparting two phases ϕ_1 and ϕ_2. Input modes are labeled by the creation operators $\hat{a}_1$ and $\hat{a}_2$ and output modes are labeled by $\hat{a}_3$ and $\hat{a}_4$.

From the quantum-information perspective discussed in Sec. II A 2, the state $|\psi\rangle$ which encodes the phase is given by:

$$|\psi\rangle = \mathrm{BS}(|1\rangle \otimes |2\rangle) \tag{27}$$

where $|i\rangle$ is the state input on the input mode i and BS imparts the transformation of the first beamsplitter on the modes. The phase encoding operation can be written as a unitary

operator, using $\phi = \phi_1 - \phi_2$ and assuming $\bar{\phi} = (\phi_1 + \phi_2)/2 = 0$ [1]:

$$U_\phi = \exp(-i\hat{n}_1\phi_1)\exp(-i\hat{n}_2\phi_2) \tag{28}$$

$$= \exp(i\hat{n}_1\phi/2)\exp(-i\hat{n}_2\phi/2) \tag{29}$$

$$= \exp(i[(\hat{n_1} - \hat{n}_2)/2]\phi). \tag{30}$$

Thus the generator of the phase encoding is proportional to the photon number difference between the arms of the interferometer: $A = (\hat{n_1} - \hat{n}_2)/2$. Considering Eq. (5), we can assert that effective states for the phase encoding should have a large variance in the photon number difference between the arms.

To understand the mapping between input modes of the interferometer and its the outputs, one can consider the following matrix formalism. The Mach-Zehnder interferometer performs a unitary operation MZ on the input $(\hat{a}_1, \hat{a}_2)$ modes, relating them to output modes $(\hat{a}_3, \hat{a}_4)$:

$$\begin{bmatrix} \hat{a}_3 \\ \hat{a}_4 \end{bmatrix} = \mathrm{MZ} \begin{bmatrix} \hat{a}_1 \\ \hat{a}_2 \end{bmatrix}. \tag{31}$$

The operator MZ comprises three unitary transformations:

$$\mathrm{MZ} = (\mathrm{BS})\Pi_\phi(\mathrm{BS}) \tag{32}$$

where BS is the action of the 50:50 beamsplitters, and Π_ϕ is the phase transformation on each mode:

$$\mathrm{BS} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \quad \Pi_\phi = \begin{bmatrix} \mathrm{e}^{i\phi/2} & 0 \\ 0 & \mathrm{e}^{-i\phi/2}, \end{bmatrix} \tag{33}$$

yielding:

$$\mathrm{MZ} = \begin{bmatrix} \cos\left(\frac{\phi}{2}\right) & i\sin\left(\frac{\phi}{2}\right) \\ i\sin\left(\frac{\phi}{2}\right) & \cos\left(\frac{\phi}{2}\right) \end{bmatrix}. \tag{34}$$

2. Homodyne Detection

Balanced homodyne detection, often called just homodyne detection, is a technique for probing a quadrature of light. Depicted in Fig. 3, homodyne detection involves mixing an unknown state $|\psi\rangle$ with a coherent state $|\beta\rangle$, called a local oscillator, on a 50:50 beamsplitter. The photon number difference is measured in the output: $\hat{M} = \hat{n}_1 - \hat{n}_2$. Using

[1] This is equivalent to the assumption that the average phase is known.

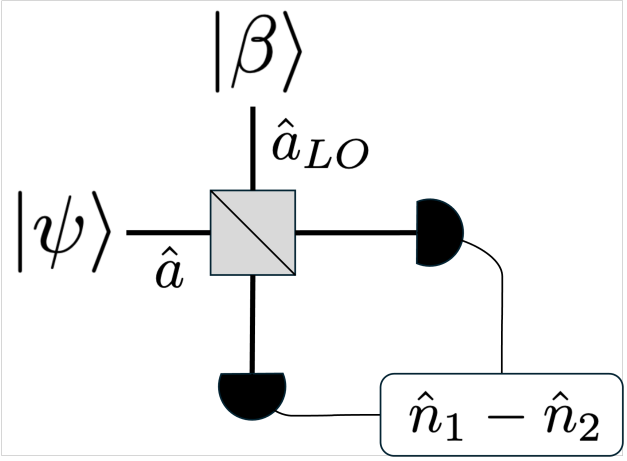


FIG. 3: Balanced homodyne detection. A state to be measured $|\psi\rangle$ is mixed with a coherent state $|\beta\rangle$ on a beamsplitter, and the photon number difference of the outputs is measured.

the beamsplitter relation, $\hat{M}$ can be written in terms of the input ladder operators for the unknown state ($\hat{a}$) and the local oscillator ($\hat{a}_{LO}$):

$$\hat{M} = \hat{a}^\dagger \hat{a}_{LO} - \hat{a}_{LO}^\dagger \hat{a} \tag{35}$$

If we treat the local oscillator as a classical state by letting $\hat{a}_{LO} = |\beta| e^{i(\theta+\pi/2)}$, an approximation made stronger for larger local oscillator amplitudes $|\beta|$ [16], then:

$$\begin{aligned} \hat{M} &= |\beta|(\hat{a}^\dagger e^{i(\theta+\pi/2)} - \hat{a} e^{-i(\theta+\pi/2)}) \\ &= i|\beta|(\hat{a}^\dagger e^{i\theta} + \hat{a} e^{-i\theta}) \\ &= 2i|\beta| x_\theta \end{aligned} \tag{36}$$

Thus for large $|\beta|$, homodyne detection approximately projects the state ψ onto a quadrature eigenstate $|x_\theta\rangle$. The probability distribution associated with the measurement is thus given by Eq. (21).

III. RESULTS

A. Phase Estimation with Caves State

Figure 4 depicts the phase estimation scheme studied. We consider the case of a symmetric Mach-Zehnder interferometer where the modes in each arm pick up a phase $\pm\frac{\phi}{2}$, implicitly assuming that the average phase between the arms remains zero. At the input

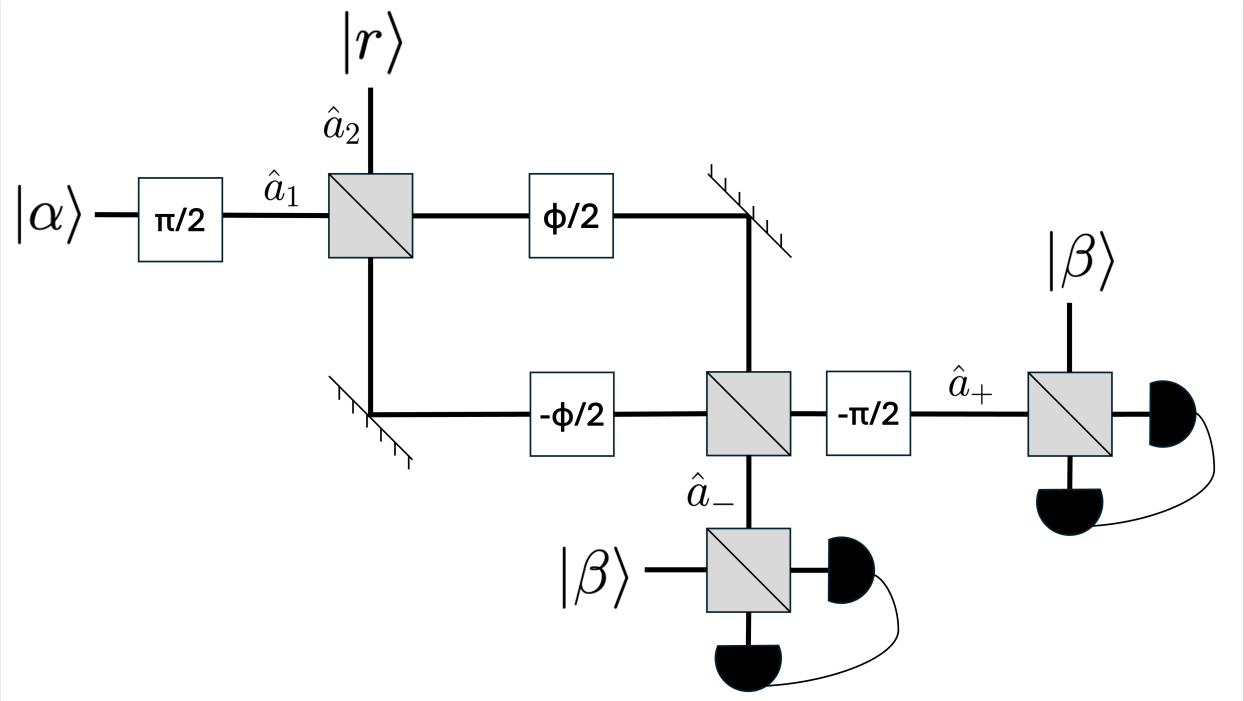


FIG. 4: Schematic of the proposed detection scheme. Coherent and squeezed-vacuum light is sent through a Mach-Zehnder interferometer, picking up a symmetric phase $\pm\phi/2$, and the output ports are each measured with homodyne detectors. Appropriate phase shifts are included at the input and output of the interferometer in order to measure the optimal quadratures.

modes a_1, a_2 we input a coherent state $|\alpha\rangle$ (with $\alpha \in \mathcal{R}$) and a squeezed vacuum state $|r\rangle$ and with a relative phase of $\pi/2$ between the modes. At each output mode a_+, a_-, the signal is probed using homodyne detection. In order to probe the correct quadrature, the output at a_+ has been shifted by a phase of $-\pi/2$ such that each local oscillator $|\beta\rangle$ is taken to be phase-locked with the input modes. This is the same detection scheme studied in [7] with an entangled coherent state instead of the coherent and squeezed light.

1. *Quantum Fisher Information*

The encoding state is given by:

$$|\psi\rangle = \mathrm{BS}(|\alpha e^{i\pi/2}\rangle \otimes |r\rangle), \tag{37}$$

known as a Caves state. Because the Caves state is a pure state, the quantum Fisher information is given by Eq. (5) with $A = (\hat{n}_1 - \hat{n}_2)/2$ as found in Sec. II C 1. This can be

shown to be [17]:

$$\alpha^2 \mathrm{e}^{2r} + \sinh^2 r. \tag{38}$$

Under the constraint that the average total photon number $\bar{n} = \alpha^2 + \sinh^2 r$ is fixed, one can show that the quantum Fisher information scales optimally with the photon number when the coherent and squeezed vacuum inputs are present with approximately the same intensity: $\bar{n}/2 \sim \alpha^2 \sim \sinh^2 r$ [17]. Operating in this optimal regime, the quantum Fisher information exhibits Heisenberg scaling [17]: $FI_Q \propto \bar{n}^2$.

2. Classical Fisher Information

To analyze the double homodyne measurement with the Caves state, we establish the unitary mapping between the input modes $(\hat{a}_1, \hat{a}_2)$ and the output modes $(\hat{a}_+, \hat{a}_-)$ of the detection scheme:

$$\begin{bmatrix} \hat{a}_+ \\ \hat{a}_- \end{bmatrix} = \begin{bmatrix} \mathrm{e}^{-i\pi/2} & 0 \\ 0 & 1 \end{bmatrix} \mathrm{MZ} \begin{bmatrix} \hat{a}_1 \\ \hat{a}_2 \end{bmatrix} \tag{39}$$

$$= \begin{bmatrix} -i\cos(\frac{\phi}{2}) & \sin(\frac{\phi}{2}) \\ i\sin(\frac{\phi}{2}) & \cos(\frac{\phi}{2}) \end{bmatrix} \begin{bmatrix} \hat{a}_1 \\ \hat{a}_2 \end{bmatrix} \tag{40}$$

Recalling that the quadrature components are the real (Eq. (15)) and imaginary (Eq. (16)) parts of the mode operators, we can use Eq. (40) to find the effect of the transformation on the quadrature operators:

$$x_1 = -p_+ \cos\left(\frac{\phi}{2}\right) + p_- \sin\left(\frac{\phi}{2}\right) \tag{41}$$

$$p_1 = x_+ \cos\left(\frac{\phi}{2}\right) - x_- \sin\left(\frac{\phi}{2}\right) \tag{42}$$

$$x_2 = x_+ \sin\left(\frac{\phi}{2}\right) + x_- \cos\left(\frac{\phi}{2}\right) \tag{43}$$

$$p_2 = p_+ \sin\left(\frac{\phi}{2}\right) + p_- \cos\left(\frac{\phi}{2}\right) \tag{44}$$

We can see that quadratures p_+, p_- are related to x_1, p_2 by a simple rotation transformation and the same is true for x_+, x_- and x_2, p_1. This fact is the result of the quadratures chosen to measure with homodyne detection (i.e. the choice of local-oscillator phases), and will be useful in straight-forwardly calculating the measurement PDF and the classical Fisher information.

With this mapping established between the quadratures, we can now consider how the phase space distribution is affected by the transformation. The two input modes are independent, so the two-mode Wigner distribution is the product of the coherent state and squeezed state Wigner functions from Eq. (25) and Eq. (26), respectively:

$$\begin{aligned} W_{\text{in}} &= W_\alpha(x_1, p_1) W_r(x_2, p_2) \\ &= \frac{4}{\pi^2} \exp\left(-2x_1^2 - 2(p_1 - \alpha)^2 - 2\mathrm{e}^{2r} x_2^2 - 2\mathrm{e}^{-2r} p_2^2\right) \end{aligned} \tag{45}$$

The mappings in Eqs. (41)-(44) allow us to find the phase space representation in terms of the output modes $W_{\text{out}}(x_+, p_+, x_-, p_-)$. The explicit expression for W_{out} is unruly, and not particularly enlightening, so we omit it here. However, the output Wigner distribution can be used to find the PDF for the double homodyne measurement by integrating over the p quadratures as in Eq. (21):

$$P(x_+, x_-) = \int_{\mathbb{R}^2} dp_+ dp_- \, W_{\text{out}}(x_+, p_+, x_-, p_-) \tag{46}$$

This gaussian integral is straightforward in the diagonalized (input) basis, so we can perform a change of basis, noting that the rotational mapping has Jacobian determinant $|J| = 1$:

$$\begin{aligned} \tilde{P}(x_2, p_1) &= \int_{\mathbb{R}^2} dx_1 dp_2 \, W_{\text{in}}(x_1, p_1, x_2, p_2) \\ &= \frac{2}{\pi} \exp\left(r - 2(p_1 - \alpha)^2 - 2\mathrm{e}^{2r} x_2^2\right) \end{aligned} \tag{47}$$

$$P(x_+, x_-) = \tilde{P}\left(x_2(x_+, x_-), p_1(x_+, x_-)\right) \tag{48}$$

Finally, we can consider the classical fisher information, given by Eq. (1), extending to the correlated measurement x_+, x_-:

$$FI_C(\phi) = \int_{\mathbb{R}^2} dx_+ dx_- \; P(x_+, x_-)^{-1} \left(\frac{\partial}{\partial \phi} P(x_+, x_-)\right)^2 \tag{49}$$

Noting that the input basis x_2, p_1 again diagonalizes the 2D gaussian, we would like to perform another change of basis. We note that $\frac{\partial P}{\partial \phi} = \frac{\partial \tilde{P}}{\partial x_2}\frac{\partial x_2}{\partial \phi} + \frac{\partial \tilde{P}}{\partial p_1}\frac{\partial p_1}{\partial \phi}$, and that $|J| = 1$ for the transformation:

$$\begin{aligned} FI_C(\phi) &= \int_{\mathbb{R}^2} dx_2 dp_1 \; (2x_2((1 - \mathrm{e}^{2r})p_1 - \alpha))^2 \tilde{P}(x_2, p_1) \\ &= \alpha^2 \mathrm{e}^{2r} + \sinh^2 r \end{aligned} \tag{50}$$

Which is identical to the quantum Fisher information in Eq. (38), meaning the double homodyne detection is an optimal measurement scheme.

B. Photon Loss

So far, we have considered an idealized case of the measurement, without considering possible experimental factors. A realistic model of the interferometer should account for some photon loss in the system, caused by random scattering of photons off the optical elements. As illustrated in Fig. 5, we modify our setup by introducing a fictional beamsplitter in each arm, each with a probability η of scattering a given photon out of the system. This introduces two more input/output modes to our consideration, which must be traced out. Because the loss modifies the encoding state, both the quantum and classical Fisher information will be changed.

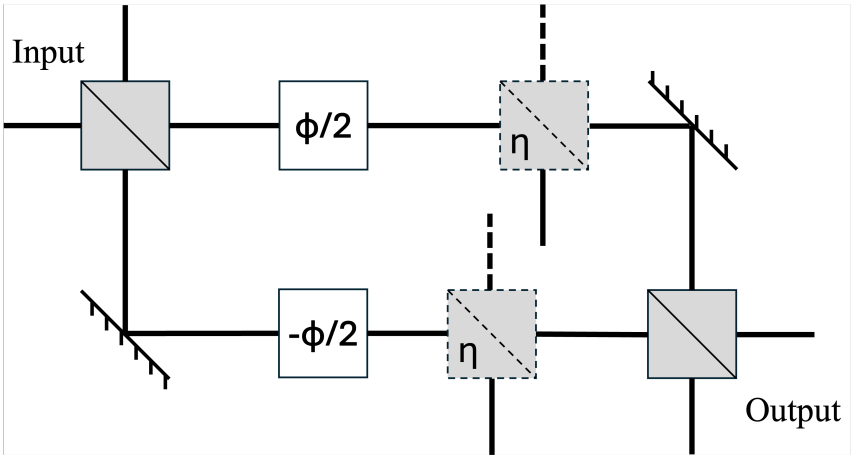


FIG. 5: Schematic of the lossy Mach-Zehnder interferometer, introducing loss in the form of beam splitters in each arm, each probability η of removing a given photon from the system. The dashed lines indicate that the vacuum is input to the beamsplitters.

1. Quantum Fisher Information

The introduction and subsequent tracing-out of the loss modes in the interferometer means that the encoded state is a mixed state, meaning our current understanding of the Quantum Fisher information as proportional to the variance with respect to the average photon number difference between the arms from Eq. (5) no longer holds. A more sophisticated treatment is required.

2. Classical Fisher Information

To account for photon losses in the interferometer, we need to incorporate the loss modes b_1, b_2, b_+, b_- into the mapping between input and output states which we found for the lossless case as in Eq. (40), as well as accounting for the fictitious beamsplitters in Fig. 5. We define the lossy Mach-Zehnder mapping:

$$\begin{bmatrix} \hat{a}_+ \\ \hat{a}_- \\ \hat{b}_+ \\ \hat{b}_- \end{bmatrix} = \begin{bmatrix} -i & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \mathrm{MZ}_\eta \begin{bmatrix} \hat{a}_1 \\ \hat{a}_2 \\ \hat{b}_1 \\ \hat{b}_2 \end{bmatrix}, \tag{51}$$

$$\mathrm{MZ}_\eta = (\mathrm{BS}) R_\eta \Pi_\phi (\mathrm{BS}), \tag{52}$$

where $\Pi_\psi, \Pi_\phi, \mathrm{BS}$ are the 4×4 extensions of the matrices defined in Eq. (33), acting with the identity on the loss modes, and R_η enacts the effect of the beamsplitters, mixing the interferometer modes with the loss modes:

$$R_\eta = \begin{bmatrix} \sqrt{1-\eta} & 0 & \sqrt{\eta} & 0 \\ 0 & \sqrt{1-\eta} & 0 & \sqrt{\eta} \\ -\sqrt{\eta} & 0 & \sqrt{1-\eta} & 0 \\ 0 & -\sqrt{\eta} & 0 & \sqrt{1-\eta} \end{bmatrix}. \tag{53}$$

Again, we consider the phase space distribution of the input and output states. The input loss modes are in the vacuum state, equivalent to a coherent state with $\alpha = 0$:

$$W_{\mathrm{vac}}(x,p) = \frac{2}{\pi} \exp\left(-2x^2 - 2p^2\right). \tag{54}$$

The total input Wigner distribution is then the product of the Wigner distributions of the input states:

$$W_{\mathrm{in}}(a_1, a_2, b_1, b_2) = W_\alpha(a_1) W_r(a_2) W_{\mathrm{vac}}(b_1) W_{\mathrm{vac}}(b_2), \tag{55}$$

where we've used the mode labels (a_i or b_i) as shorthand to denote both x and p quadrature for that mode. The output distribution $W_{\mathrm{out}}(a_+, a_-, b_+, b_-)$ is found by replacing the mode variables according to Eq. (51). Since the photons which exit the interferometer cannot be measured, the state which is accessible to be measured is the partial trace with respect to the loss modes:

$$\tilde{W}_{\mathrm{out}}(x_+, p_+, x_-, p_-) = \int_{\mathcal{R}^4} d^2 b_+ d^2 b_- \, W_{\mathrm{out}}(a_+, a_-, b_+, b_-), \tag{56}$$

and the PDF for the double homodyne measurement is:

$$\begin{aligned} P(x_+, x_-) &= \int_{\mathcal{R}^2} dp_+ dp_- \, \tilde{W}_{\text{out}}(x_+, p_+, x_-, p_-) \\ &= \int_{\mathcal{R}^6} dp_+ dp_- d^2b_+ d^2b_- \, W_{\text{out}}(a_+, a_-, b_+, b_-). \end{aligned} \tag{57}$$

The introduction of the loss modes makes the diagonalization process less straightforward than in the lossless case. To evaluate the integral in Eq. (57), we have to reduce the problem from 8 dimensions (4 modes, each with 2 quadratures) to one in 6 dimensions, tracing out the loss modes and the p quadratures in the measurement modes. Define the vectors of integration variables $\mathbf{y} = (p_+, p_-, x_{b_+}, p_{b_+}, x_{b_-}, p_{b_-})$ and measurement variables $\mathbf{x} = (x_+, x_-)$. By transforming Eq. (55) via Eq. (52), W_{out} will have the form:

$$W_{\text{out}}(\mathbf{x}, \mathbf{y}) = \frac{16}{\pi^4} \exp\left(f(\mathbf{x}, \mathbf{y})\right), \tag{58}$$

where f is a bilinear function in the quadrature variables. We can thus solve for $A \in \mathcal{R}^{6\times 6}$, $\mathbf{B}(\mathbf{x}) : \mathcal{R}^2 \to \mathcal{R}^6$ and $c(\mathbf{x}) : \mathcal{R}^2 \to \mathcal{R}$ such that:

$$f(\mathbf{x}, \mathbf{y}) = \mathbf{y} \cdot A \cdot \mathbf{y} + \mathbf{B}(\mathbf{x}) \cdot \mathbf{y} + c(\mathbf{x}). \tag{59}$$

The PDF Eq. (57) is then given by the known integral [18]:

$$\begin{aligned} P(x_+, x_-) &= \frac{16}{\pi^4} \int_{\mathcal{R}^6} d^6\mathbf{y} \, \exp\left(\mathbf{y} \cdot A \cdot \mathbf{y} + \mathbf{B}(\mathbf{x}) \cdot \mathbf{y} + c(\mathbf{x})\right) \\ &= \frac{16}{\pi^4} \sqrt{\frac{(2\pi)^6}{\det A}} \exp\left(\frac{1}{2}\mathbf{B}(\mathbf{x}) \cdot A^{-1} \cdot \mathbf{B}(\mathbf{x}) + c(\mathbf{x})\right). \end{aligned} \tag{60}$$

Note that in general, A, $\mathbf{B}$ and $c(\mathbf{x})$ depend on ϕ. This procedure yields the following PDF for the lossy case:

$$\tilde{P}(z_+, z_-) = \frac{2}{\pi} \sqrt{\chi_\eta} \exp(-2\chi_\eta z_-^2 - 2(z_+ - \alpha\sqrt{1-\eta})^2), \tag{61}$$

where:

$$\begin{bmatrix} z_+ \\ z_- \end{bmatrix} = \begin{bmatrix} \cos\left(\frac{\phi}{2}\right) & -\sin\left(\frac{\phi}{2}\right) \\ \sin\left(\frac{\phi}{2}\right) & \cos\left(\frac{\phi}{2}\right) \end{bmatrix} \begin{bmatrix} x_+ \\ x_- \end{bmatrix},$$

$$\chi_\eta = \frac{1}{(1-\eta)\mathrm{e}^{-2r} + \eta}.$$

Noting that the gaussian PDF is diagonalized in the rotated basis of z_+, z_-, we can perform a change of basis to calculate the classical fisher information via Eq. (1), which yields:

$$FI_C(\phi) = \frac{(1-\eta)}{1 + 2\eta\mathrm{e}^r \sinh r} (\alpha^2 \mathrm{e}^{2r} + (1-\eta)\sinh^2 r). \tag{62}$$

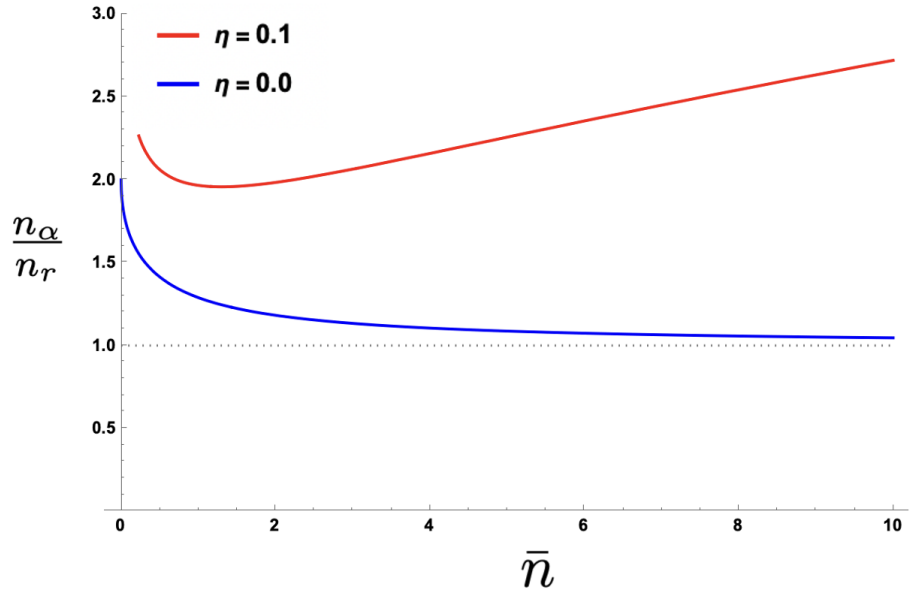


FIG. 6: Optimal ratio between intensity of input coherent light $n_\alpha = \alpha^2$ and squeezed vacuum light $n_r = \sinh^2 r$ as a function of the average photon number $\bar{n} = n_\alpha + n_r$, for the cases of no loss ($\eta = 0$) and 10% loss ($\eta = 0.1$).

To operate in the optimal regime for fixed average photon number $\bar{n} = \alpha^2 + \sinh^2 r$, we optimize using Lagrange multipliers, solving the system of equations:

$$\frac{\partial FI_C}{\partial \alpha} = \lambda \frac{\partial \bar{n}}{\partial \alpha}, \tag{63}$$

$$\frac{\partial FI_C}{\partial r} = \lambda \frac{\partial \bar{n}}{\partial r}, \tag{64}$$

where λ is a Lagrange multiplier. For fixed photon loss η, the above system can be solved numerically to find the ratio of input coherent light $n_\alpha = \alpha^2$ to squeezed vacuum light $n_r = \sinh^2 r$. We found that in general $n_\alpha > n_r$ for non-zero loss, in contrast to the $\eta = 0$ case where $n_\alpha \; n_r$ is optimal [17].

IV. DISCUSSION

A. Optimal Measurement Scheme with Heisenberg Scaling

In Sec. III A 2, we found that the classical Fisher information for the proposed double homodyne detection scheme is equal to the quantum Fisher information, meaning it is an optimal measurement scheme for a Caves state. To our knowledge, this fact is not well

known. A very recent study [19] has considered a double-homodyne detection scheme of the Caves state for the simultaneous estimation of both the phase difference ϕ and the average phase $\bar{\phi}$ between the arms (taken to be zero here). From their work, the result from Eq. (50) can be straightforwardly computed, though it was not explicitly stated.

This optimal Fisher information scales with the square of the average photon number (Heisenberg scaling), which makes it a highly sensitive measurement, especially useful in applications where intensity of light is limited. Such Heisenberg-scaling detection schemes exist, though many incur practical difficulties in state generation and/or measurement stages. Compared with alternative encoding states (e.g. NOON states [10], entangled coherent states [7], two-mode squeezed states [20]), the Caves state is comparatively easy to prepare. Furthermore, homodyne detection is a tried-and-true measurement with few requirements for sophisticated equipment, compared to alternative measurements (e.g. photon-number resolving detectors [21], photon-number parity measurements [22]). The proposed measurement scheme is therefore an excellent candidate for phase estimation.

B. Sensitivity to Photon Loss

In Sec. III B 2 we considered the effect of photon loss in the proposed phase estimation scheme. Our results found that the Caves state is particularly robust to photon loss. Firstly, from Eq. (62), we note that the information from the measurement of the lossy state is independent of the phase being measured, meaning that the measurement scheme need not be updated during measurement, which is not necessarily true for optimal lossy detection schemes. Secondly, the proposed measurement retains the Heisenberg scaling for a reasonable amount of photon loss.

Fig. 7 compares the normalized phase uncertainty as a function of average photon number with various losses $\eta = 0, 0.01, 0.1$ as a function of the average photon number. The phase uncertainty is normalized with respect to the standard quantum limit (SQL) which is achieved by a classical interferometer, and the optimal ratio of input coherent/squeezed light was determined by solving Eqs. (63), (64). A similar plot appears in the study of the ECS [7], for the ECS state. Around $\bar{n} = 10$, corresponding to 1 photon lost, the advantage of the ECS state compared with SQL is lost. We attribute this to the high degree of path entanglement in the ECS state: the lost photon behaves as measurement on the path degree-of-freedom, collapsing the wavefunction and losing phase information about one of the interferometer arms. In comparison, the Caves state does not exhibit this behaviour,

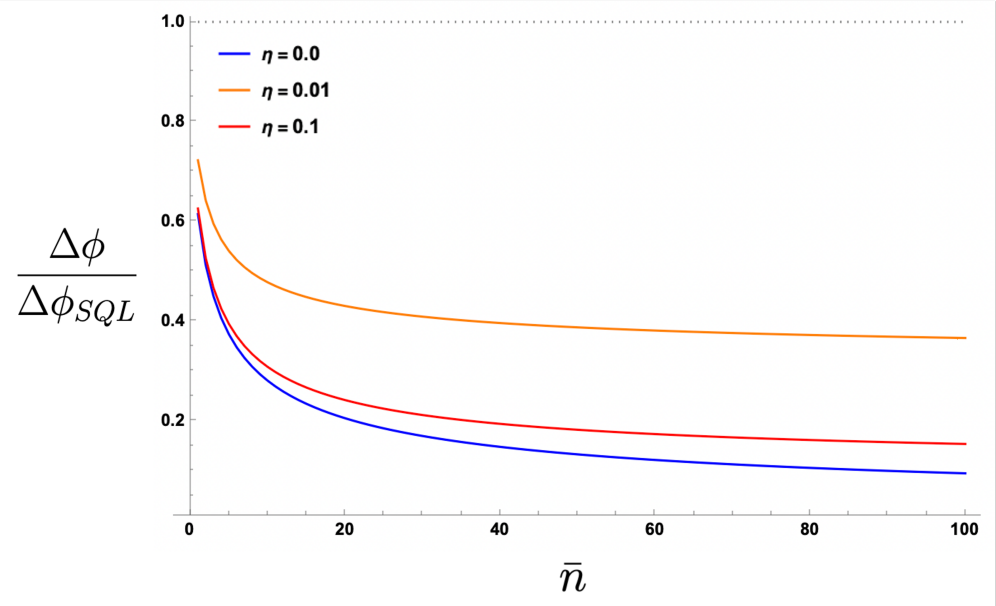


FIG. 7: Optimal phase uncertainty $\Delta\phi$ normalized with respect to the standard quantum limit $\Delta\phi_{SQL} = [(1-\eta)\bar{n}]^{-1/2}$ for various loss factors, as a function of the average photon number.

'beating' the photon loss in the limit of large $\bar{n}$.

C. Further Considerations

This work has been an effort towards an eventual experimental realization of sub-SQL phase estimation. To this end, we've considered the effect of photon loss on the detection scheme. There are, however, more considerations to be made for a practical implementation. Firstly, though the classical Fisher information retains its Heisenberg scaling in the presence of reasonable photon loss, we have not yet performed a calculation of the quantum Fisher information in this case. If the detection scheme is non-optimal, we could perhaps modify the detection scheme to achieve optimality. We may also consider other models of noise in a practical implementation, namely the presence of anti-squeezed light in the interferometer. Significantly higher squeezing could be achieved practically at the cost of producing a mixed state of squeezed and anti-squeezed light in the interferometer. We likely need to compare the cost/benefit of pushing the squeezing to higher values of r.

V. CONCLUSIONS

This project considered the phase estimation problem in a Mach-Zehnder interferometer for the case where the average phase difference is known. We have proposed a double-homodyne detection scheme using squeezed vacuum and coherent light and demonstrated that it saturated the quantum Cramér-Rao bound in the ideal (lossless) case. Furthermore, we considered the effect of photon loss on the proposed detection scheme and found that the classical Fisher information is independent of the true value of the phase being measured, and remains increasing for increased average photon number with fixed loss. These results in combination with the comparative practicality of the estimation scheme make the proposed measurement an excellent candidate for experimental realization of sub-SQL phase-estimation.

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